Projective Gauge Reduction and Auxiliary Connection Elimination in a Metric-Affine Trace Sector

Abstract

Independent affine connections are routinely eliminated algebraically in Palatini and metric-affine gravity, but projective gauge degeneracy makes it nontrivial to determine whether elimination defines a unique reduced theory. A local Hessian criterion separates gauge freedom from invariant auxiliary degeneracy. In a four-dimensional trace/vector model with the compensator A_μ varied, the full pointwise one-form-fibre Hessian kernel equals the projective gauge image precisely on the regular locus $f_1 \neq 0$, $f_2 \neq 0$. The horizontal solution is unique, the affine solutions form one local orbit, and quotient-first and eliminate-first reductions agree as local bulk functionals modulo the admitted divergence currents. At $f_1 = 0$ or $f_2 = 0$, rank loss is not merely enlargement of projective gauge freedom: source compatibility may fail, while compatible branches retain gauge-invariant algebraic moduli. These statements are local to the stated trace/vector configuration space.

1 Introduction

When may a gauge-degenerate auxiliary field be eliminated before or after taking the gauge quotient without changing the reduced theory?

An independent affine connection in Palatini and metric-affine gravity is typically auxiliary, and the Palatini curvature scalar is invariant under a consistent projective shift [1–6]. Neither that projective invariance nor algebraic connection elimination is claimed as new [7–12]. What is not automatic is that the two operations agree.

Gauge nonuniqueness and auxiliary nonuniqueness need not coincide. Raw affine nonuniqueness splits into two distinct effects. Some directions are gauge: they never enter the bulk density and are removed by the projective quotient. Other directions are gauge-invariant and may nevertheless remain auxiliary. Gauge invariance is not dynamical independence, and gauge redundancy is not the same as auxiliary eliminability. Only after those two effects are separated is auxiliary elimination guaranteed to define a reduced spectator theory rather than a representative choice among invariant degeneracies.

Independent affine connections are commonly eliminated algebraically in Palatini and metric-affine analyses [4, 5, 7–11, 13]. If the affine Hessian has gauge-invariant null directions beyond the projective orbit, that substitution does not define a unique quotient theory: unresolved invariant data remain. The exact kernel criterion is therefore a local diagnostic of when integrating out the connection is a quotient-independent operation rather than an implicit non-gauge branch restriction. The diagnostic is established here in the stated trace/vector sector; it is not claimed for unrestricted metric-affine gravity.

The trace/vector sector is a minimal nontrivial setting in this model where a projective one-form orbit, gauge-invariant horizontal affine data, and rank-changing algebraic connection equations coexist. It is therefore an exact laboratory for separating projective gauge degeneracy from invariant failure of auxiliary eliminability. This defined configuration space is not asserted to be a dynamically consistent truncation of unrestricted 64-component metric-affine gravity.

On the regular locus of the model below, the vertical projective direction is quotiented and the horizontal affine variables are uniquely eliminated. Beyond it, extra gauge-invariant kernel can leave unresolved invariant data rather than additional projective freedom. The algebraic source conditions are sharp: S2 ($f_1 = 0$, $f_2 \neq 0$) requires $\partial f_1 = 0$; S3 ($f_1 \neq 0$, $f_2 = 0$) requires $\partial f_3 = 0$; and S4 ($f_1 = f_2 = 0$) requires both. Incompatible sources give no stationary affine solution, while compatible branches can retain invariant moduli. These statements classify algebraic branches, not dynamical crossings.

The existence of a projectively invariant residue is not new here. A Palatini torsion construction realized projective invariance through a scalar spurion ϵ, held fixed at the stated order, with invariant combination $T_\mu - \partial_\mu \epsilon$ [14]. A subsequent operator-basis analysis used an independent one-form A_μ as a nondynamical reference on the projective fibre, again with residue $T_\mu - A_\mu$, without imposing $\delta S/\delta A$ [15]. The present paper addresses a distinct variational and reduction question: A_μ is included in the auxiliary variational system; the full affine Hessian is computed and its kernel is compared with the projective tangent; local orbit uniqueness is established on the regular locus; and quotient-first and eliminate-first reductions are compared explicitly. Varying A_μ is not proposed as the unique physical completion of the earlier nondynamical construction; it defines the distinct auxiliary variational problem studied here. In that problem, full-Hessian rank, source compatibility, orbit uniqueness, and elimination order are tested together. That a strict projection need not agree with a Schur/Feshbach reduction is a general methodological point [16]; it is used only as a reason not to assume that reduction order is innocuous, not as metric-affine evidence.

Section 2 records a finite-dimensional organizing criterion for quadratic auxiliary densities. It is a reduction lemma, not a deep new general mathematical theorem. Sections 3–6 supply the principal scientific content: a nontrivial verification of its hypotheses in a four-dimensional metric-affine trace/vector model. The Palatini kinetic density $3(\partial f_1)^2/(2f_1)$ is generic structure [4, 7, 8]. In this varied-A_μ system, the model-specific contribution is the full affine Hessian and its exact full-fibre gauge kernel, the sharp regular-locus local orbit, explicit equality of quotient-first and eliminate-first reductions, and singular-stratum source compatibility. The transverse determinant $-f_1^2 f_2/6$ supplies the regularity test for those results.

2 A finite-dimensional reduction criterion

Let E be a finite-dimensional auxiliary fibre over spectator fields ψ, with coordinates X^A and bulk density

$$L_{\rm aux}(X;\psi) = \tfrac{1}{2} X^A H_{AB}(\psi) X^B + J_A(\psi) X^A + L_0(\psi), \tag{2.1}$$

H symmetric. Let $G : \mathfrak{g} \to E$ be linear and injective, acting only in the fibre: $\delta_\varepsilon X = G\varepsilon$.

Gauge invariance $L_{\rm aux}(X + G\varepsilon) = L_{\rm aux}(X)$ for all X, ε expands to

$$HG = 0, \qquad G^T J = 0. \tag{2.2}$$

No further algebraic condition follows from invariance of (2.1).

Theorem 2.1 (Finite-dimensional gauge-auxiliary reduction criterion)**.** *Assume* (2.2) *and* $\ker H = \operatorname{im} G$. *Equivalently, in finite dimensions, the induced map* $\overline{H} : E/\operatorname{im} G \to (E/\operatorname{im} G)^*$ *is invertible. Then:*

1. $HX + J = 0$ *has a unique solution class in* $E/\operatorname{im} G$*;*
2. *the unreduced solution set is one affine gauge orbit* $X_\star + \operatorname{im} G$*;*
3. *the on-shell value* $L_{\rm on} = L_0 + \frac{1}{2} J^T X$ *is constant on that orbit;*

4. *quotient-first elimination and eliminate-first quotient produce the same local on-shell functional;*

5. *any two linear horizontal charts related by* $\mathrm{GL}(n)$ *have the same regular locus and the same on-shell value.*

Proof. Invariance gives $\operatorname{im} G \subseteq \ker H$; the kernel hypothesis gives equality. Solvability of $HX = -J$ is $J \in (\operatorname{im} G)^0$, which is $G^T J = 0$. The solution space is one affine orbit of direction $\ker H$. Finite-dimensional nondegeneracy of $\overline{H}$ yields a unique class $[X_\star]$. On shell, $L_{\mathrm{on}} = L_0 + \frac{1}{2} J^T X$, and $J^T G = 0$ makes this representative-independent. Both reduction orders substitute the same class into (2.1). A $\mathrm{GL}(n)$ change of horizontal coordinates multiplies the determinant of the induced transverse form $H_\perp$ by a nowhere-zero squared Jacobian, leaving the regular locus and the on-shell scalar L_{on} invariant. □

Theorem 2.1 provides an organizing finite-dimensional criterion for a quadratic form with a prescribed kernel. Sufficiency is all that is used below. It does not replace a model Hessian computation.

In a splitting $E = \mathcal{H} \oplus \operatorname{im} G$ with coordinates (x, V),

$$H = \begin{pmatrix} H_\perp & 0 \\ 0 & 0 \end{pmatrix}, \tag{2.3}$$

and $\ker H = \operatorname{span}\{e_V\}$ precisely when $\det H_\perp \neq 0$, with $x_\star = -H_\perp^{-1} J_{\mathcal{H}}$ and $L_{\mathrm{on}} = L_0 - \frac{1}{2} J_{\mathcal{H}}^T H_\perp^{-1} J_{\mathcal{H}}$.

Remark 2.2 (Species matrices and the one-form fibre). If the auxiliary fibre at a spacetime point x is of the form $E_x = S \otimes T_x^* M$ and the quadratic density is built from a species matrix H_{AB} contracted with a nondegenerate spectator metric, the true Hessian is $H \otimes g^{-1}$. Then

$$\ker(H \otimes g^{-1}) = \ker H \otimes T_x^* M, \qquad \operatorname{rank}(H \otimes g^{-1}) = D \operatorname{rank}_{\mathrm{sp}}(H). \tag{2.4}$$

Species invertibility and kernel criteria therefore lift directly to the full one-form fibre. Theorem 2.1 is applied below to that full pointwise fibre; the displayed 3×3 and 4×4 matrices are the species coefficients.

Uniqueness of the reduced solve for *every* compatible source requires $\ker H = \operatorname{im} G$. If $\ker H$ strictly contains $\operatorname{im} G$, the quotient retains additional gauge-invariant degeneracies; selecting a representative of those directions requires a non-gauge restriction, branch choice, or additional dynamics, and is not a gauge quotient. Equality of on-shell *values* for a single compatible source can still occur; unrestricted necessity of QE = EQ is not claimed. The criterion is fibrewise linear algebra. It is not a symplectic reduction theorem [12, 17, 18].

3 Metric-affine trace model

Spacetime dimension $D = 4$, signature $(-,+,+,+)$. The curvature, torsion, and nonmetricity conventions are

$$R^\lambda{}_{\rho\mu\nu} = \partial_\mu \Gamma^\lambda{}_{\nu\rho} - \partial_\nu \Gamma^\lambda{}_{\mu\rho} + \Gamma^\lambda{}_{\mu\sigma}\Gamma^\sigma{}_{\nu\rho} - \Gamma^\lambda{}_{\nu\sigma}\Gamma^\sigma{}_{\mu\rho}, \tag{3.1}$$

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}, \qquad R = g^{\mu\nu} R_{\mu\nu}, \tag{3.2}$$

$$T^\lambda{}_{\mu\nu} = 2\Gamma^\lambda{}_{[\mu\nu]}, \qquad T_\mu = T^\lambda{}_{\lambda\mu}, \tag{3.3}$$

$$Q_{\lambda\mu\nu} = -\nabla_\lambda g_{\mu\nu}, \qquad Q_\lambda = g^{\mu\nu} Q_{\lambda\mu\nu}, \qquad \widetilde{Q}_\nu = g^{\mu\lambda} Q_{\lambda\mu\nu}. \tag{3.4}$$

The affine derivative on the metric is

$$\nabla_\lambda g_{\mu\nu} = \partial_\lambda g_{\mu\nu} - \Gamma^\rho{}_{\lambda\mu} g_{\rho\nu} - \Gamma^\rho{}_{\lambda\nu} g_{\mu\rho}. \tag{3.5}$$

The affine configuration space of the model is the three traces $(T_\mu, Q_\mu, \widetilde{Q}_\mu)$ together with a compensator one-form A_μ. Axial torsion, tensor torsion, and traceless nonmetricity exist as metric-affine irreps [1, 19] and are not coordinates here: they are neither solved to zero nor removed by the projective gauge. Matter is metric-minimal, so no hypermomentum sources the connection [20, 21].

Assumption 3.1 (Trace/vector configuration space)**.** The independent affine data are $(T, Q, \widetilde{Q}, A)$. Deleted irreps remain outside the domain.

Assumption 3.2 (Metric-minimal matter)**.** Matter does not supply an independent hypermomentum source for the connection equation.

$$\bar{T}_\mu := T_\mu - A_\mu. \tag{3.6}$$

The residue $t_\mu := \bar{T}_\mu$ is the projectively invariant torsion-trace combination. The bulk action is

$$S = \int \mathrm{d}^4x\, \sqrt{-g}\, \Big[f_1\, R(\Gamma) + f_2\, \bar{T}^2 + f_3\, \bar{\nabla} \cdot \bar{T} + U(\phi) \Big], \tag{3.7}$$

with coefficient functions of a scalar argument ϕ. This is a representative, compact, nontrivial projectively invariant trace-residue model: $f_1 R(\Gamma)$ supplies the projective gauge and the Palatini block, while $f_2 \bar{T}^2 + f_3 \bar{\nabla} \cdot \bar{T}$ keep a gauge-invariant affine residue algebraically active. It is not claimed to be the most general metric-affine Lagrangian, nor a unique minimal operator basis.

Assumption 3.3 (Trace-residue action)**.** The bulk functional is (3.7). The potential $U(\phi)$ is affine-independent.

The one-form A_μ is introduced as a compensator field for the one-form projective action. No microscopic or ultraviolet Stueckelberg interpretation is assumed. The combination T^2 is not projectively invariant, while $\bar{T}^2$ is, once $\delta A = 3\xi$ accompanies the connection shift. In the gauge-adapted coordinates of Section 4, $V_\mu = A_\mu$ parametrizes the projective orbit, while $t_\mu = T_\mu - A_\mu$ is gauge invariant but remains auxiliary on U_{reg}. The field A_μ is varied as an auxiliary field in (3.7); after the quotient it parametrizes the orbit rather than an independent bulk degree of freedom.

4 Projective transformation and invariant coordinates

The projective transformation used throughout is

$$\delta\Gamma^\lambda{}_{\mu\nu} = -\delta^\lambda{}_\nu \xi_\mu, \qquad \delta A_\mu = 3\xi_\mu, \tag{4.1}$$

with $\delta g_{\mu\nu} = 0$. Index placement is part of the law under the targets below. The MAG first-kind shift $\delta\Gamma^\lambda{}_{\mu\nu} = +\delta^\lambda{}_\mu \xi_\nu$ [1, 6] is a different transformation, not used here.

Linear vectorial connection maps form a larger family [6, 22]. The two-parameter subfamily containing common first- and last-index writings [2, 3, 23] is

$$\delta\Gamma^\lambda{}_{\mu\nu} = a\, \delta^\lambda{}_\mu \xi_\nu + b\, \delta^\lambda{}_\nu \xi_\mu, \tag{4.2}$$

with compensator increment held at

$$\delta A_\mu = +3\xi_\mu. \tag{4.3}$$

At $D = 4$,

$$\delta T_\mu = 3(a-b)\xi_\mu, \qquad \delta Q_\mu = (2a+8b)\xi_\mu, \qquad \delta\widetilde{Q}_\mu = (5a+2b)\xi_\mu. \tag{4.4}$$

The linear Ricci shift is

$$\delta R_{\alpha\beta}\big|_{\partial} = b\,\partial_\alpha\xi_\beta + (-3a-b)\,\partial_\beta\xi_\alpha. \tag{4.5}$$

The quadratic remainder in the connection difference vanishes if and only if $a = 0$. The Ricci target

$$\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}. \tag{4.6}$$

together with $\delta T = +3\xi$ forces

$$a = 0, \qquad b = -1. \tag{4.7}$$

Proposition 4.1 (Selection within the adopted two-parameter connection-shift family)**.** *Under the conventions of Section 3 and the simultaneous targets*

$$\delta T = +3\xi, \qquad \delta A = +3\xi, \qquad \delta(T-A) = 0, \qquad \delta R = 0, \qquad \delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}, \tag{4.8}$$

the family (4.2) *admits the unique solution* $a = 0$, $b = -1$. *Equivalently, the law is* (4.1). *A constant rescaling of* ξ *reparametrizes the same law.*

The induced shifts are

$$\delta(T_\mu, Q_\mu, \widetilde{Q}_\mu, A_\mu) = (3, -8, -2, 3)\xi_\mu. \tag{4.9}$$

Hence $\delta\bar{T} = 0$ and

$$\delta R^{\lambda}{}_{\rho\mu\nu} = -\delta^{\lambda}{}_{\rho}\, 2\partial_{[\mu}\xi_{\nu]}, \tag{4.10}$$

so $\delta R = 0$. The first-kind pair $(a, b) = (1, 0)$ yields

$$\delta(T, Q, \widetilde{Q}) = (3, 2, 5)\xi \tag{4.11}$$

and fails (4.6); it is excluded only under these targets.

Write

$$\Gamma^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}_{\mathrm{LC}\,\mu\nu} + N^{\lambda}{}_{\mu\nu}. \tag{4.12}$$

The trace-sector distortion is

$$\begin{aligned} N^{\lambda}{}_{\mu\nu} = {} & \tfrac{1}{3}\big(\delta^{\lambda}{}_{\mu}T_\nu - g_{\mu\nu}T^{\lambda}\big) \\ & + \tfrac{1}{36}\big(5\delta^{\lambda}{}_{\mu}Q_\nu + 5\delta^{\lambda}{}_{\nu}Q_\mu - 7g_{\mu\nu}Q^{\lambda}\big) \\ & + \tfrac{1}{18}\big(-\delta^{\lambda}{}_{\mu}\widetilde{Q}_\nu - \delta^{\lambda}{}_{\nu}\widetilde{Q}_\mu + 5g_{\mu\nu}\widetilde{Q}^{\lambda}\big), \end{aligned} \tag{4.13}$$

Invariant coordinates adapted to (4.1) are

$$t_\mu = T_\mu - A_\mu, \qquad q_\mu = Q_\mu + \tfrac{8}{3}A_\mu, \qquad u_\mu = \widetilde{Q}_\mu + \tfrac{2}{3}A_\mu, \qquad V_\mu = A_\mu, \tag{4.14}$$

with inverse

$$T = t + V, \qquad Q = q - \tfrac{8}{3}V, \qquad \widetilde{Q} = u - \tfrac{2}{3}V, \qquad A = V. \tag{4.15}$$

Thus $t = \bar{T}$.

Lemma 4.2 (Horizontal invariance)**.** *Under* (4.9), $\delta t = \delta q = \delta u = 0$ *and* $\delta V = 3\xi$. *The coordinate* $V_\mu = A_\mu$ *is vertical;* (q, u, t) *are invariant.*

Assumption 4.3 (Regular locus)**.** Invertibility, orbit, and reduced-action statements are made on

$$U_{\mathrm{reg}} := \{f_1 \neq 0\} \cap \{f_2 \neq 0\}. \tag{4.16}$$

Assumption 4.4 (Local bulk identification)**.** The theorem is spacetime-local on open regular regions. Bulk densities are identified modulo the admitted total divergences generated by the currents J_1 and J_3 of Appendix H. That identification is a variational-boundary convention. It does not classify proper versus improper projective transformations.

Signs $f_1 > 0$, $f_2 > 0$, and $K_{\mathrm{eff}} > 0$ are not hypotheses.

Table 1: Conventions and projective transformations at $D = 4$.

Object	Definition or law	Variation
$R^{\lambda}{}_{\rho\mu\nu}$	(3.1)	$-\delta^{\lambda}{}_{\rho}\, 2\partial_{[\mu}\xi_{\nu]}$
$R_{\mu\nu}$	(3.2)	$-2\partial_{[\mu}\xi_{\nu]}$
R	$g^{\mu\nu}R_{\mu\nu}$	0
T_μ	(3.3)	$+3\xi_\mu$
Q_μ	(3.4)	$-8\xi_\mu$
$\widetilde{Q}_\mu$	(3.4)	$-2\xi_\mu$
A_μ	compensator	$+3\xi_\mu$
$\bar{T}_\mu$	$T_\mu - A_\mu$	0
$\Gamma^{\lambda}{}_{\mu\nu}$	independent connection	$-\delta^{\lambda}{}_{\nu}\xi_\mu$

5 Exact transverse Hessian

5.1 Exact trace expansion

Insert (4.13) into the connection-difference identity for (3.1) and contract to the scalar curvature. Every quadratic and divergence coefficient is determined. There is no remainder in the trace sector:

$$\begin{aligned} R - R_{\mathrm{LC}} = \bar{\nabla}_\mu\big(-2T^\mu - Q^\mu + \widetilde{Q}^\mu\big) \\ - \tfrac{2}{3}T^2 - \tfrac{11}{72}Q^2 + \tfrac{1}{18}\widetilde{Q}^2 - \tfrac{2}{3}T\cdot Q + \tfrac{2}{3}T\cdot\widetilde{Q} + \tfrac{2}{9}Q\cdot\widetilde{Q}. \end{aligned} \tag{5.1}$$

Here $X\cdot Y := g^{\mu\nu}X_\mu Y_\nu$ and $X^2 := X\cdot X$. The six quadratic coefficients are

$$\Big(-\tfrac{2}{3},\ -\tfrac{11}{72},\ +\tfrac{1}{18},\ -\tfrac{2}{3},\ +\tfrac{2}{3},\ +\tfrac{2}{9}\Big) \tag{5.2}$$

in the order $(T^2, Q^2, \widetilde{Q}^2, T\cdot Q, T\cdot\widetilde{Q}, Q\cdot\widetilde{Q})$. Equation (5.1) is the complete expansion used below; no ellipsis is implied. The derivation is given in Appendix C.

5.2 Projective cancellation in raw traces

Substituting the projective shifts (4.9) into the divergence and into the quadratic polynomial of (5.1) cancels all vertical dependence. Equivalently, after (4.15) both the divergence current

$$W^\mu := -2T^\mu - Q^\mu + \widetilde{Q}^\mu \tag{5.3}$$

and the quadratic form become functions of (t, q, u) only:

$$W^\mu = -2t^\mu - q^\mu + u^\mu, \tag{5.4}$$

and the quadratic polynomial is that of (5.1) with $(T, Q, \widetilde{Q})$ replaced by (t, q, u). A constant pure vertical connection has vanishing scalar curvature, which is an independent control of $\delta R = 0$.

5.3 Integration by parts

Multiply (5.1) by f_1 and use the two bulk identities

$$\sqrt{-g}\, f_1\, \bar{\nabla}_\mu W^\mu = -\sqrt{-g}\,(\partial_\mu f_1)W^\mu + \partial_\mu\big(\sqrt{-g}\, f_1 W^\mu\big), \tag{5.5}$$

$$\sqrt{-g}\, f_3\, \bar{\nabla}_\mu \bar{T}^\mu = -\sqrt{-g}\,(\partial_\mu f_3)\bar{T}^\mu + \partial_\mu\big(\sqrt{-g}\, f_3 \bar{T}^\mu\big). \tag{5.6}$$

The currents actually generated are

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 \bar{T}^\mu. \tag{5.7}$$

Bulk functionals that differ by $\partial_\mu\big(\sqrt{-g}\,(c_1 J_1^\mu + c_3 J_3^\mu)\big)$ are identified. That is the admitted variational-boundary class; it is not a classification of all possible surface terms, and it does not distinguish proper from improper projective transformations (Appendix H).

5.4 Horizontal affine quadratic form

After (5.5)–(5.6) and the horizontal rewrite, the affine bulk density (omitting $f_1 R_{\rm LC} + U(\phi)$ and omitting the admitted divergence) is quadratic-plus-linear in the species triple $x := (q, u, t)^{\mathsf{T}}$:

$$\mathcal{L}_{\rm aff} = \tfrac{1}{2} x^{\mathsf{T}} H_\perp x + J^{\mathsf{T}} x. \tag{5.8}$$

Each species coordinate is a one-form. The displayed 3×3 matrix $H_\perp$ is a *species-space* coefficient matrix: the quadratic pairing is the metric contraction $X^A \cdot Y^B := g^{\mu\nu} X^A_\mu Y^B_\nu$. At each spacetime point x write $S_\perp \simeq \mathbb{R}^3$ and

$$E_{\perp,x} = S_\perp \otimes T^*_x M. \tag{5.9}$$

The true pointwise transverse Hessian is therefore

$$\mathcal{H}_\perp = H_\perp \otimes g^{-1}. \tag{5.10}$$

Because g^{-1} is nondegenerate,

$$\ker \mathcal{H}_\perp = \ker H_\perp \otimes T^*_x M, \qquad \operatorname{rank} \mathcal{H}_\perp = D \operatorname{rank}_{\rm sp}(H_\perp). \tag{5.11}$$

The species matrix and source are

$$H_\perp = f_1 H_0 + \operatorname{diag}(0, 0, 2f_2), \tag{5.12}$$

$$H_0 = \begin{pmatrix} -11/36 & 2/9 & -2/3 \\ 2/9 & 1/9 & 2/3 \\ -2/3 & 2/3 & -4/3 \end{pmatrix}, \tag{5.13}$$

$$J = \big(\partial f_1,\ -\partial f_1,\ 2\partial f_1 - \partial f_3\big)^{\mathsf{T}}. \tag{5.14}$$

There is no V dependence. The matrix H_0 is the Hessian of the quadratic polynomial in (5.1) with respect to (q, u, t). The diagonal $2f_2$ is the Hessian contribution of $f_2 t^2$. The source (5.14) is the linear form generated by (5.5)–(5.6). Detailed contractions are in Appendices C and D.

The two nonmetricity traces (q, u) enter $H_\perp$ through the upper-left 2×2 block. Isolating that block is the Palatini step: it eliminates the connection data that mix with f_1 before the residue equation is solved. The existence of such a Palatini block is generic [4, 5, 8]; the numerical matrix below is convention-specific.

5.5 The exact matrix

Write $q_N := (q, u)^{\mathsf{T}}$. Then

$$P := f_1 \begin{pmatrix} -11/36 & 2/9 \\ 2/9 & 1/9 \end{pmatrix}. \tag{5.15}$$

A direct determinant computation gives

$$\det P = f_1^2\Big(\big(-\tfrac{11}{36}\big)\big(\tfrac{1}{9}\big) - \big(\tfrac{2}{9}\big)^2\Big) = f_1^2\Big(-\tfrac{11}{324} - \tfrac{4}{81}\Big) = f_1^2\Big(-\tfrac{11}{324} - \tfrac{16}{324}\Big) = -\frac{f_1^2}{12}. \tag{5.16}$$

Hence P is invertible if and only if $f_1 \neq 0$. The inverse is

$$P^{-1} = f_1^{-1} \begin{pmatrix} -4/3 & 8/3 \\ 8/3 & 11/3 \end{pmatrix}. \tag{5.17}$$

No further field-dependent factor occurs in $\det P$. In particular the schematic notations M_{Pal} and Palatini_response are replaced, for all subsequent statements, by (5.15)–(5.17).

Lemma 5.1 (Palatini invertibility)**.** *On the present domain,* $\mathrm{rank}_{\mathrm{sp}}(P) = 2$ *if and only if* $f_1 \neq 0$*. At* $f_1 = 0$ *one has* $P = 0$*, so the Palatini pair cannot be solved.*

5.6 Explicit q/u solve

At fixed residue t, the (q, u) Euler–Lagrange equation is

$$P\, q_N + f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix} t + \begin{pmatrix} \partial f_1 \\ -\partial f_1 \end{pmatrix} = 0. \tag{5.18}$$

Applying (5.17) yields

$$q = 4\,\frac{\partial f_1}{f_1} - \frac{8}{3}t, \qquad u = \frac{\partial f_1}{f_1} - \frac{2}{3}t, \tag{5.19}$$

on $f_1 \neq 0$. The algebra is recorded in Appendix D. Equation (5.19) is the complete Palatini response in the horizontal chart: there is no unspecified matrix standing between the source and (q, u).

The same inverse applied to the f_1-sourced part alone produces the Palatini particular solution

$$(q, u)\big|_{t=0} = \big(4r,\, r\big), \qquad r := \frac{\partial f_1}{f_1}, \tag{5.20}$$

which will combine with the residue solution in the next section.

With (q, u) eliminated by (5.19), the remaining horizontal unknown is the invariant residue $t = \bar{T}$. The reduced t-problem is the Schur complement of P in $H_\perp$.

5.7 Exact Schur data

A direct complement computation (Appendix E) produces

$$S_t = 2f_2, \qquad J_{t,\mathrm{eff}} = -\partial f_3, \qquad \mathcal{L}_{\mathrm{const}} = \frac{3(\partial f_1)^2}{2f_1}. \tag{5.21}$$

The on-shell Palatini contribution is therefore the constant (in t) density $3(\partial f_1)^2/(2f_1)$, and the remaining quadratic-plus-linear t-density is

$$f_2\, t^2 - (\partial f_3) \cdot t. \tag{5.22}$$

5.8 Vanishing of the induced residue couplings

Write the would-be induced couplings after Palatini elimination as

$$\widehat{f_2} = f_2 + \Delta_2[f_1], \qquad \widehat{f_3} = f_3 + \Delta_3[f_1]. \tag{5.23}$$

Comparing (5.21)–(5.22) with the original residue terms in (3.7) gives

$$\Delta_2[f_1] = 0, \qquad \Delta_3[f_1] = 0, \tag{5.24}$$

hence

$$\widehat{f_2} = f_2, \qquad \widehat{f_3} = f_3. \tag{5.25}$$

These zeros are exact results of the Schur complement, not assumptions and not artifacts of the canonical (q, u, t) chart. Four independent evaluations—raw Euler–Lagrange substitution, the Schur complement, an alternate GL(3) horizontal basis, and substitution of the complete raw solution—all give $\Delta_2 = \Delta_3 = 0$ and the same on-shell density.

5.9 Residue equation

Stationarity of (5.22) is

$$2f_2\, t^\mu - \partial^\mu f_3 = 0. \tag{5.26}$$

On $f_2 \neq 0$,

$$t_\star = \frac{\partial f_3}{2f_2}. \tag{5.27}$$

Combining (5.19) with (5.27) and writing

$$r := \frac{\partial f_1}{f_1}, \qquad s := \frac{\partial f_3}{2f_2}, \tag{5.28}$$

one obtains the complete horizontal solution

$$t = s, \qquad q = 4r - \tfrac{8}{3}s, \qquad u = r - \tfrac{2}{3}s. \tag{5.29}$$

Equivalently,

$$q_\star = 4\frac{\partial f_1}{f_1} - \frac{4}{3}\frac{\partial f_3}{f_2}, \qquad u_\star = \frac{\partial f_1}{f_1} - \frac{1}{3}\frac{\partial f_3}{f_2}. \tag{5.30}$$

No step in (5.21)–(5.29) extends through $f_1 = 0$ or $f_2 = 0$.

6 Application: kernel, orbit, and commutativity

6.1 Transverse determinant

Because $H_\perp$ is block-Schur with Palatini block P and complement $S_t = 2f_2$,

$$\det H_\perp = (\det P)\,(2f_2) = -\frac{f_1^2 f_2}{6}. \tag{6.1}$$

Therefore the species matrix $H_\perp$ is invertible if and only if $f_1 \neq 0$ and $f_2 \neq 0$. On that locus $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 3$. By (5.11) the pointwise one-form-fibre rank is then $\operatorname{rank}\mathcal{H}_\perp = D \cdot 3$, hence 12 at $D = 4$.

6.2 Full affine Hessian

Restore the vertical coordinate V. Write $S_{\mathrm{full}} \simeq \mathbb{R}^4$ for the species space of (q, u, t, V) and

$$E_{\mathrm{full},x} = S_{\mathrm{full}} \otimes T_x^* M. \tag{6.2}$$

The bulk density (5.8) is independent of V, so the species Hessian in (q, u, t, V) is the 4×4 coefficient matrix

$$H_{\mathrm{full}} = \begin{pmatrix} H_\perp & 0 \\ 0 & 0 \end{pmatrix}. \tag{6.3}$$

The true pointwise Hessian on the full one-form fibre is

$$\mathcal{H}_{\mathrm{full}} = H_{\mathrm{full}} \otimes g^{-1}. \tag{6.4}$$

Because g^{-1} is nondegenerate,

$$\ker \mathcal{H}_{\mathrm{full}} = \ker H_{\mathrm{full}} \otimes T_x^* M, \qquad \operatorname{rank}\mathcal{H}_{\mathrm{full}} = D \operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}). \tag{6.5}$$

On U_{reg} one has $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$ and $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$. At $D = 4$ the corresponding fibre multiplicities are $\operatorname{rank}\mathcal{H}_{\mathrm{full}} = 12$ and $\dim\ker\mathcal{H}_{\mathrm{full}} = 4$. Species inclusion is

$$\ker H_{\mathrm{full}} \supseteq \operatorname{span}\{e_V\}, \qquad e_V = (0, 0, 0, 1). \tag{6.6}$$

Define the pointwise projective gauge map

$$G_x : T_x^*M \to E_{\mathrm{full},x}, \qquad \xi_\mu \mapsto e_V \otimes (3\xi_\mu). \tag{6.7}$$

The factor 3 is the vertical shift $\delta V = 3\xi$ of (4.1); any nonzero scalar yields the same image. Thus

$$\operatorname{im} G_x = \operatorname{span}\{e_V\} \otimes T_x^*M. \tag{6.8}$$

Proposition 6.1 (Exact kernel)**.** *On U_{reg}, the species kernel is*

$$\ker H_{\mathrm{full}} = \operatorname{span}\{e_V\}, \tag{6.9}$$

and the true pointwise kernel on the full one-form fibre is

$$\ker \mathcal{H}_{\mathrm{full}} = \operatorname{im} G_x. \tag{6.10}$$

Equivalently, in raw traces the species kernel is $\ker H_{\mathrm{raw}} = \operatorname{span}\{(3,-8,-2,3)\}$*, and the raw full-fibre kernel is*

$$\big\{(3,-8,-2,3)\,\xi_\mu : \xi \in T_x^*M\big\}. \tag{6.11}$$

This is a fibre-level restatement of the species result, not a new algebraic theorem. The projective gauge image exhausts the affine degeneracy of (3.7) *in the present trace sector: there is no additional Hessian kernel.*

Proof. Inclusion $\operatorname{span}\{e_V\} \subseteq \ker H_{\mathrm{full}}$ is (6.3). For the converse, if $H_{\mathrm{full}}(x,V) = 0$ with $x = (q,u,t)$, then $H_\perp x = 0$. On U_{reg}, $\det H_\perp \neq 0$, so $x = 0$, while V remains arbitrary. Thus $\ker H_{\mathrm{full}} \subseteq \operatorname{span}\{e_V\}$. Tensoring with T_x^*M and using (6.5)–(6.8) yields (6.10).

Independently, the raw bulk density after the admitted integrations by parts is quadratic in $(T, Q, \widetilde{Q}, A)$ and annihilates the projective species vector $(3,-8,-2,3)$ identically (Appendix F). On U_{reg} that species kernel is one-dimensional, so the fibre kernel is D-dimensional. The coordinate change (4.15) identifies it with $\operatorname{im} G_x$. The vertical Euler–Lagrange combination

$$\mathrm{EL}_T - \tfrac{8}{3}\mathrm{EL}_Q - \tfrac{2}{3}\mathrm{EL}_{\widetilde{Q}} + \mathrm{EL}_A \tag{6.12}$$

vanishes identically, as required of a gauge identity. □

6.3 Complete connection solution

Translating (5.29) back to raw traces, with V arbitrary,

$$\begin{aligned} T &= s + V, \\ Q &= 4r - \tfrac{8}{3}(s+V), \\ \widetilde{Q} &= r - \tfrac{2}{3}(s+V), \\ A &= V. \end{aligned} \tag{6.13}$$

The reconstructed distortion on this solution is

$$N^\lambda{}_{\mu\nu} = \tfrac{1}{2}\big(\delta^\lambda{}_\mu r_\nu + \delta^\lambda{}_\nu r_\mu - g_{\mu\nu} r^\lambda\big) - \tfrac{1}{3}\delta^\lambda{}_\nu (s_\mu + V_\mu). \tag{6.14}$$

Shifting the representative by $V \mapsto V + 3\xi$ changes Γ by exactly $-\delta^\lambda{}_\nu \xi_\mu$, which is (4.1).

Theorem 6.2 (Local connection orbit)**.** *Assume Assumptions 3.1–4.4 and the law* (4.1)*. On U_{reg}, at fixed admissible (g,ϕ):*

1. *there is one pointwise algebraic solution of the trace-sector affine Euler–Lagrange equations, unique up to the raw projective vector* $(3,-8,-2,3)$*;*

2. *if the coefficient functions and the regularity conditions are smooth on a spacetime-local regular region, these pointwise solutions assemble into one spacetime-local orbit modulo the projective transformation* (4.1).

The explicit solution is (6.13)–(6.14). *Smoothness of the coefficient functions on a regular region is the regularity actually used; no Banach-manifold quotient or global slice is invoked. The statement does not classify Hamiltonian or covariant-phase-space charges of ξ, and is not a statement about deleted affine irreps.*

The qualifier "the connection solution" is therefore to be read as this local orbit, never as a globally unique connection.

6.4 The two routes

Let U_{reg} be as in (4.16), and let bulk densities be identified modulo the admitted class (5.7).

Route QE **(quotient first).** Replace the raw traces $(T, Q, \widetilde{Q}, A)$ by the invariant horizontal coordinates (t, q, u), discarding V. Solve the nonsingular three-species system $H_{\perp}x + J = 0$ on U_{reg}. Substitute the unique horizontal solution (5.29) into (5.8).

Route EQ **(eliminate first).** Retain an arbitrary representative V. Substitute the complete raw solution (6.13) into the original trace-expanded bulk density, before any horizontal reduction.

These are not merely names of a diagram: each is an explicit substitution, written in Appendix G. In local field theory, substitution of some fields by local functionals of others commutes with the Euler–Lagrange problem when those fields are auxiliary, and need not commute otherwise [12]. That general fact is the conceptual setting of QE = EQ. It is not a substitute for the Hessian computation of Section 6 or for the substitutions of Appendix G.

6.5 Equality of on-shell densities

Theorem 6.3 (Local bulk quotient–elimination commutativity)**.** *On U_{reg}, as spacetime-local bulk functionals modulo the displayed total divergences generated by J_1 and J_3,*

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \mathcal{L}^{\mathrm{EQ}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}. \tag{6.15}$$

The eliminate-first density is manifestly independent of V. Equality is a bulk identification. It does not classify projective charges, does not extend through singular branches, and does not supply a global gauge slice.

Proof. The substitutions are written in Appendix G. On U_{reg} the horizontal system has the unique solution (5.29). Substituting into (5.8) produces the right-hand side of (6.15) by the Schur evaluation (5.21) together with (5.27); this is QE. For EQ, substitute (6.13) into the raw quadratic-plus-linear density (F.1). Every term involving V cancels by projective invariance of that density (Proposition 6.1), and the remaining expression coincides with (6.15). The two substitutions therefore agree as bulk densities. Identification uses the admitted currents (5.7). Whether those total divergences are retained, canceled, or discarded is a choice of variational boundary data, independent of any proper/improper classification of ξ (Appendix H). □

The same bulk value is obtained for the representatives $V = 0$, $V = \mathrm{d}\chi$, and V an arbitrary one-form. Representative independence of the bulk is therefore part of (6.15), not an extra assumption.

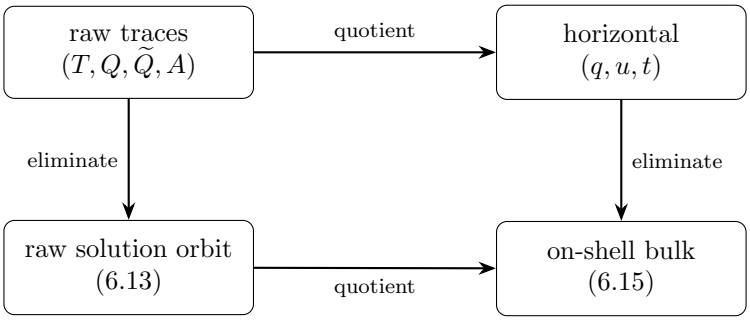


Figure 1: Commutative diagram for Theorem 6.3. Both arrows to the on-shell bulk density are explicit substitutions (Appendix G), equal as spacetime-local bulk functionals modulo the admitted IBP currents J_1, J_3, on U_{reg}.

6.6 Where the local and IBP hypotheses enter

Manifest V-independence of the eliminate-first bulk density follows from exact projective invariance, equivalently the explicit absence of V in the horizontal density. On $f_1 f_2 \neq 0$, $\ker \mathcal{H}_{\mathrm{full}} = \operatorname{im} G_x$ excludes additional gauge-invariant null directions and makes the horizontal quotient solve unique. Identification of the two bulk functionals uses (5.5)–(5.6) and discards $\partial_\mu(\sqrt{-g}(c_1 J_1^\mu + c_3 J_3^\mu))$ as a variational-boundary convention. Passage from a pointwise algebraic orbit to a spacetime-local orbit uses smoothness of (f_i, V) on a regular region. None of these steps is a global slice, and none classifies projective charges.

7 Reduced action

7.1 Bulk functional

Collecting $f_1 R_{\mathrm{LC}}$, the on-shell affine density (6.15), and the affine-independent potential, one obtains

$$S_{\mathrm{red}} = \int \mathrm{d}^4x\, \sqrt{-g}\left[f_1\, R_{\mathrm{LC}} + \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2} + U(\phi)\right] + \text{admitted boundary}. \tag{7.1}$$

The boundary addendum is of the class (5.7). Equation (7.1) is a spacetime-local bulk functional on U_{reg}, not a complete initial-boundary-value action.

7.2 Coefficient functions of a scalar

If $f_i = f_i(\phi)$ and $U = U(\phi)$, then $\partial_\mu f_i = f_i'\, \partial_\mu \phi$ and

$$K_{\mathrm{Pal}}(\phi) = \frac{3(f_1')^2}{2f_1}, \tag{7.2}$$

$$K_{\mathrm{eff}}(\phi) = \frac{3(f_1')^2}{2f_1} - \frac{(f_3')^2}{4f_2}. \tag{7.3}$$

The reduced bulk Lagrangian then reads $f_1 R_{\mathrm{LC}} + K_{\mathrm{eff}}\,(\partial\phi)^2 + U(\phi)$. The writing

$$\frac{3}{2}\frac{(\partial f_1)^2}{f_1} \tag{7.4}$$

is the same Palatini density as (7.2). The schematic ratio-square

$$\frac{3}{2}\Big(\frac{f_1'}{f_1}\Big)^2 \tag{7.5}$$

is not equivalent: it differs from (7.2) by a factor $1/f_1$ and is not used. The coefficient $3(f_1')^2/(2f_1)$ is the generic Palatini kinetic contribution associated with eliminating an independent connection in $f_1 R(\Gamma)$ [4, 7, 8]. Its presence here is reproduced structure, not a new Palatini identity.

Proposition 7.1 (Kinetic identification)**.** *The Palatini contribution generated by eliminating (q, u) is exactly* (7.2)*, equivalently* (7.4)*. The residue contribution generated by* (5.27) *is $-(f_3')^2/(4f_2)$. Their sum is K_{eff}. In particular*

$$K_{\mathrm{eff}} \neq Z_{\mathrm{geom}} \tag{7.6}$$

as a theorem statement. Here Z_{geom} stands only for any geometric constitutive kinetic factor that one might wish to identify with the reduced coefficient; no such factor is supplied by the affine linear algebra, and none is identified with K_{eff}.

7.3 Conditional scalar role

If ϕ is varied as a dynamical field, the reduced scalar Euler–Lagrange equation is that of (7.1). If ϕ is a background argument of the coefficient functions, no scalar Cauchy problem is asserted. The two modes are not mixed. Auxiliary elimination does not, by itself, generate a constitutive predictivity theorem: it produces (7.1) on U_{reg}, and nothing further about the observational content of K_{eff}.

7.4 Main theorem

Theorem 7.2 (Exact local projective-affine reduction)**.** *Assume the trace/vector domain of Section 3, the projective law* (4.1)*, metric-minimal matter, and the admitted IBP class of Assumption 4.4. On $U_{\mathrm{reg}} = \{f_1 \neq 0\} \cap \{f_2 \neq 0\}$:*

1. *the affine Hessian kernel is exactly the projective tangent, and the transverse Hessian is invertible;*
2. *affine solutions form one local orbit in the sense of Theorem 6.2;*
3. *quotient and auxiliary elimination commute in the sense of Theorem 6.3;*
4. *the reduced bulk functional is* (7.1)*, with* (7.2)–(7.3) *when $f_i = f_i(\phi)$.*

The scalar role remains conditional. Signs $f_1 > 0$, $f_2 > 0$, and $K_{\mathrm{eff}} > 0$ are not hypotheses. Deleted irreps remain outside the domain.

8 Singular strata and limitations

8.1 Four algebraic strata

The vanishing of $\det H_\perp = -f_1^2 f_2/6$ defines two hypersurfaces in coefficient space. Together they partition the problem into four strata. The horizontal system is

$$H_\perp x + J = 0, \qquad x = (q, u, t)^{\mathsf{T}}, \tag{8.1}$$

with J as in (5.14). The classification below uses only this linear algebra: no division by f_1 or f_2 is performed until invertibility has been established.
At $D = 4$ the fibre multiplicities are therefore S1: $\operatorname{rank} \mathcal{H}_{\mathrm{full}} = 12$, $\dim \ker \mathcal{H}_{\mathrm{full}} = 4$; S2: 4 and 12; S3: 8 and 8; S4: 0 and 16. Source-compatibility statements are unchanged.

Table 2: Regular versus singular strata of the affine Hessian, including source compatibility of (8.1). Rank and nullity columns are species-space values of the coefficient matrices. In $D = 4$ the corresponding pointwise one-form-fibre ranks and nullities are obtained by multiplication by four because $g^{\mu\nu}$ is nondegenerate. The reduced action of Theorem 7.2 exists only on S1. The vertical coordinate V remains a projective gauge parameter on every compatible stratum.

	condition	$\mathrm{rank}_{\mathrm{sp}} P$	$\mathrm{rank}_{\mathrm{sp}} H_\perp$	$\mathrm{null}_{\mathrm{sp}} H_{\mathrm{full}}$	source / existence	invariant moduli	Thm. 7.2
S1	$f_1 \neq 0,\ f_2 \neq 0$	2	3	1	always solvable; unique x	none	yes
S2	$f_1 = 0,\ f_2 \neq 0$	0	1	3	$\partial f_1 = 0$; else none	if compatible: q, u; $t = \partial f_3/(2f_2)$	no
S3	$f_1 \neq 0,\ f_2 = 0$	2	2	2	$\partial f_3 = 0$; else none	if compatible: t; q, u of t	no
S4	$f_1 = 0,\ f_2 = 0$	0	0	4	$\partial f_1 = \partial f_3 = 0$; else none	if compatible: q, u, t	no

On S1 the species kernel is exactly vertical, equivalently $\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x$, and the reduced action (7.1) exists. On S2, compatibility requires $\partial f_1 = 0$; if that holds, $t = \partial f_3/(2f_2)$ while q and u remain invariant algebraic moduli, whereas $\partial f_1 \neq 0$ yields no stationary horizontal solution. On S3, compatibility requires $\partial f_3 = 0$; if that holds, t is a gauge-invariant modulus and (q, u) are determined as functions of t, whereas $\partial f_3 \neq 0$ yields no stationary solution. On S4, compatibility requires $\partial f_1 = \partial f_3 = 0$; if that holds, (q, u, t) are horizontal moduli, whereas otherwise there is no stationary solution. Extra kernel on S2–S4 consists of these invariant degeneracies: it is not additional projective gauge. No additional algebraic factor of $\det H_\perp$ exists, and no regular proof step extends through S2–S4. Those branches are not solved dynamically here.

Figure 2 maps the four regions.

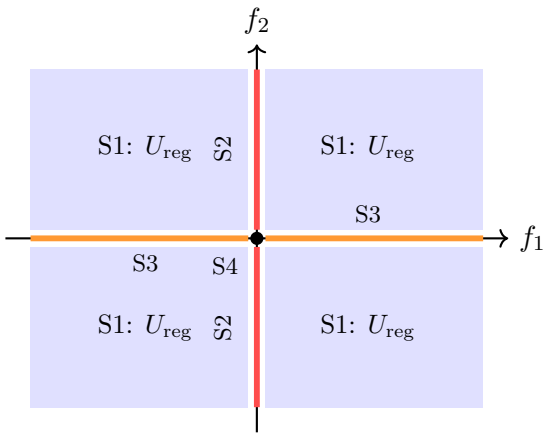


Figure 2: Regular branch S1 versus singular strata S2–S4 in the complete (f_1, f_2) coefficient plane. Algebraic regularity is $f_1 \neq 0$ and $f_2 \neq 0$: S1 comprises all four open quadrants. The vertical axis excluding the origin is S2; the horizontal axis excluding the origin is S3; the origin is S4. No sign condition is imposed by the algebraic regularity theorem. Axes themselves are excluded from Theorem 7.2.

8.2 Necessary counterexamples

The hypotheses are sharp in the following sense (Appendix I).

- If $f_1 = 0$, Palatini invertibility fails, and $\partial f_1 \neq 0$ makes (8.1) inconsistent.
- If $f_2 = 0$, residue invertibility fails, and $\partial f_3 \neq 0$ makes (8.1) inconsistent.
- If the projective law is replaced by $(a, b) = (1, 0)$, the Ricci target (4.6) fails.
- If deleted irreps are treated as solved Hessian coordinates, the domain premise fails.

8.3 Claim boundary

The following remain outside Theorem 7.2.

Table 3: Scope and excluded sectors. Excluded irreducible sectors lie outside the stated configuration space; they are neither proved to vanish nor removed by projective gauge symmetry.

Object or claim	Classification
axial torsion	excluded by truncation
tensor torsion	excluded by truncation
traceless nonmetricity	excluded by truncation
higher affine irreps	excluded by truncation
hypermomentum	not covered
independent matter connection sources	not covered
global gauge slice	not claimed
Hamiltonian / covariant-phase-space charges of ξ	not analyzed
boundary-charge algebra	not classified
extension through $f_1 = 0$ or $f_2 = 0$	not claimed
mandatory scalar dynamics	conditional only
$f_1 > 0$, $f_2 > 0$, $K_{\text{eff}} > 0$	not hypotheses
ghost-freedom / energy stability	not claimed
$K_{\text{eff}} = Z_{\text{geom}}$	identification not established
constitutive predictivity	not claimed
unrestricted metric-affine (64-component) theorem	not claimed

In particular, algebraic regularity is $f_1 \neq 0$ and $f_2 \neq 0$. Physical-stability inequalities are a different and absent theorem. Whether the displayed IBP divergences are retained as boundary data is a variational convention, not a charge classification.

9 Discussion

9.1 Gauge versus auxiliary nonuniqueness

Raw traces $(T, Q, \widetilde{Q}, A)$ split as $(q, u, t \mid V)$. The vertical coordinate $V_\mu = A_\mu$ parametrizes the projective orbit. The invariants (q, u, t) are not gauge-removable, yet on U_{reg} they are algebraic. In particular $t = T - A$ is projectively invariant and is solved by $t_\star = \partial f_3/(2f_2)$. Gauge-invariant does not mean propagating. After elimination, $f_i(\phi)$ and $U(\phi)$ remain constitutive data: auxiliary linear algebra does not fix them and does not identify K_{eff} with a geometric factor Z_{geom}.

Theorem 6.3 is the finite-dimensional criterion 2.1 plus the evaluation $\ker \mathcal{H}_{\text{full}} = \operatorname{im} G_x$ on U_{reg}, equivalently $\ker H_{\text{full}} = \operatorname{span}\{e_V\}$ in species space. Unrestricted necessity of commutativity is not claimed: on-shell *values* can agree for a special compatible source even if extra kernel

is present. Unique reduced elimination for every compatible source does require $\ker H = \operatorname{im} G$. Extra kernel, when it occurs, consists of gauge-invariant degeneracies.

9.2 Fingerprint $\Delta_2 = \Delta_3 = 0$

Palatini elimination does not renormalize the residue couplings: $\Delta_2 = \Delta_3 = 0$. Appendix J tests that fingerprint against the two-parameter quadratic deformation family (J.2), without altering the defining equations of (3.7). The induced correction to the t-Schur coefficient is generically nonzero; both corrections vanish simultaneously only at $(\alpha, \beta) = (0, 0)$. An independent linear-source deformation is a separate supporting control and is not part of the proposition below. Thus $\Delta_2 = \Delta_3 = 0$ is a structural fingerprint of (3.7) within the tested family, not a claim about all conceivable symmetry-compatible operators.

Proposition 9.1 (Non-genericity of the residue fingerprint)**.** *Within the two-parameter family* (J.2)*, $\Delta_2 = \Delta_3 = 0$ holds simultaneously only at the undeformed point. The zeros are therefore classified as explicitly non-generic inside that family.*

9.3 Singular crossings of $f_i(\phi)$

If $f_i = f_i(\phi)$, a locus $f_1(\phi_*) = 0$ has $\partial f_1 = f_1'(\phi_*)\,\mathrm{d}\phi$. A generic crossing with $f_1'(\phi_*) \neq 0$ and $\mathrm{d}\phi \neq 0$ therefore violates the S2 compatibility condition $\partial f_1 = 0$ and yields no stationary affine solution. Only a source-compatible crossing can support the S2 moduli branch. On $f_2(\phi_*) = 0$, compatibility is controlled by $\partial f_3 = 0$, not by ∂f_2. This is an algebraic reading of Table 2; it is not a claim of dynamical singularity propagation or of cosmological pathology.

9.4 Relation to metric-affine literature

Projective Palatini gravity and Ricci-based MAG connection solutions are known [2, 3, 6, 9, 23]. Palatini $f(R)$ and the kinetic shift $3(F')^2/(2F)$ are known [4, 5, 7, 8]. Projectively invariant scalar-metric-affine models integrate the connection out to DHOST-type theories [10, 24]. Quadratic MAG constructions often solve the connection after breaking or otherwise controlling projective degeneracy [11, 13]. Unified Palatini treatments with torsion or nonmetricity give precise conditions for propagating versus algebraic distortion [25]. Auxiliary substitution commutativity [12] is the conceptual setting, not a substitute MAG proof. Table 4 records a positive-scope comparison of these sources. The related residue-lineage and reduction-order sources are discussed in the Introduction rather than in this MAG table.

Recent scalar-metric-affine work likewise eliminates an algebraic independent connection on a generic nondegenerate branch and encodes its response in an effective scalar kinetic function [26]. This application motivates the complementary structural question here: when is such elimination unique modulo projective gauge?

The condition $\ker H = \operatorname{im} G$ is the general finite-dimensional auxiliary diagnostic used here. By contrast, $\det H_\perp = -f_1^2 f_2/6$ and the exact realization of the projective gauge kernel hold for the stated trace/vector configuration space. Restoring omitted affine irreducible sectors would require a new Hessian analysis; this sector is not established as a dynamically consistent truncation of unrestricted 64-component MAG [1, 19].

9.5 Auxiliary elimination is not predictivity

Solving Γ and A produces (7.1). Ghost-freedom, $f_1 > 0$, and observational content are outside the theorem. Global slices and Hamiltonian charges of projective transformations remain open [2].

Table 4: Positive-scope comparison of related metric-affine and substitution sources. Entries record established content in each work's own setting. Related residue and reduction-order sources are discussed in the Introduction.

Work	Affine setting	Principal reduction/connection result	Relation to the present question
Dadhich–Pons [3]	Palatini Einstein–Hilbert	Projective orbit of the Palatini connection; Levi-Civita plus projective mode	Projective equivalence of Palatini gravity; independent of the present Hessian test
Alfonso et al. [9]	Ricci-based MAG	Algebraic connection solution with projective invariance retained; torsion role trivialized in the observables	Ricci-based algebraic Γ solve; distinct configuration space
Aoki–Shimada [10, 24]	Scalar metric-affine	Connection integrated out to a projectively protected scalar-metric theory	Projective ghost protection after Γ elimination; different operator content
Iosifidis 2019 [11]	Solvable MAG	Explicit algebraic connection solutions under stated solvability criteria	Algebraic Γ solvability in a broader MAG class
Iosifidis 2022 [13]	Quadratic MAG	Quadratic connection solve under nondegeneracy hypotheses, often with controlled projective data	Unique representative after nondegeneracy hypotheses
Böhmer–Jensko [25]	Unified Palatini MAG	Criteria distinguishing algebraic versus propagating distortion	Distortion-propagation tests in Palatini MAG
Pons 2010 [12]	Local field substitution	Conditions under which substituting fields in the action commutes with the Euler–Lagrange problem	Conceptual setting of QE = EQ; not a MAG model
present work	Trace/vector residue $T - A$	Exact projective kernel on the full one-form fibre; unique local orbit and QE = EQ on U_{reg}	Species determinant $\det H_\perp$ and source-compatible singular strata

10 Conclusion

Gauge nonuniqueness and auxiliary nonuniqueness need not coincide. For a local quadratic auxiliary density, invariance along a gauge distribution together with $\ker H = \operatorname{im} G$ is a sufficient finite-dimensional reduction criterion: it yields a unique reduced solve, one affine gauge orbit, and equality of quotient-first and eliminate-first on-shell functionals. That organizing lemma is not the principal claim. In the metric-affine trace/vector model (3.7), the curvature and compensator targets select (4.1) within the adopted two-parameter family. On $f_1 \neq 0$ and $f_2 \neq 0$ the species determinant is $\det H_\perp = -f_1^2 f_2/6$, so the pointwise one-form-fibre kernel is exactly the projective gauge image. Affine solutions form one local orbit, QE = EQ holds as spacetime-local bulk functionals modulo the displayed currents J_1 and J_3, and the reduced functional is (7.1) with the standard Palatini kinetic term $3(f_1')^2/(2f_1)$.

On S2 ($f_1 = 0$, $f_2 \neq 0$), stationary affine solutions require $\partial f_1 = 0$; on S3 ($f_1 \neq 0$, $f_2 = 0$), they require $\partial f_3 = 0$; and on S4 ($f_1 = f_2 = 0$), both conditions are required. Rank loss can thus mean source incompatibility or gauge-invariant algebraic moduli, not merely extra projective gauge freedom. These conclusions are local to the trace/vector configuration space;

they do not extend the regular reduced action through the singular strata or address projective charges, stability, or constitutive predictivity.

A Conventions and projective transformation

This appendix reconstructs Table 1 and Proposition 4.1.

A.1 Index conventions

$D = 4$, signature $(-,+,+,+)$. $A_{(\mu\nu)} = (A_{\mu\nu} + A_{\nu\mu})/2$ and $A_{[\mu\nu]} = (A_{\mu\nu} - A_{\nu\mu})/2$. The curvature, Ricci, torsion, and nonmetricity definitions are (3.1)–(3.5).

A.2 Trace map of the two-parameter family

Let $\delta\Gamma^{\lambda}{}_{\mu\nu} = a\delta^{\lambda}{}_{\mu}\xi_{\nu} + b\delta^{\lambda}{}_{\nu}\xi_{\mu}$ and $\delta g_{\mu\nu} = 0$. Then

$$\delta T^{\lambda}{}_{\mu\nu} = (a-b)\big(\delta^{\lambda}{}_{\mu}\xi_{\nu} - \delta^{\lambda}{}_{\nu}\xi_{\mu}\big), \tag{A.1}$$

hence

$$\delta T_{\mu} = \delta T^{\lambda}{}_{\lambda\mu} = (a-b)\big(D\xi_{\mu} - \xi_{\mu}\big) = (D-1)(a-b)\xi_{\mu}. \tag{A.2}$$

At $D = 4$ this is $\delta T = 3(a-b)\xi$. For nonmetricity, $Q_{\lambda\mu\nu} = -\nabla_{\lambda} g_{\mu\nu}$ and $\delta g = 0$ give

$$\delta Q_{\lambda\mu\nu} = \delta\Gamma^{\rho}{}_{\lambda\mu} g_{\rho\nu} + \delta\Gamma^{\rho}{}_{\lambda\nu} g_{\mu\rho}. \tag{A.3}$$

With $\delta\Gamma^{\rho}{}_{\lambda\mu} = a\delta^{\rho}{}_{\lambda}\xi_{\mu} + b\delta^{\rho}{}_{\mu}\xi_{\lambda}$,

$$\delta Q_{\lambda\mu\nu} = a\big(g_{\lambda\nu}\xi_{\mu} + g_{\lambda\mu}\xi_{\nu}\big) + 2b\, g_{\mu\nu}\xi_{\lambda}. \tag{A.4}$$

Contracting,

$$\delta Q_{\lambda} = g^{\mu\nu}\delta Q_{\lambda\mu\nu} = (2a + 2Db)\xi_{\lambda}, \tag{A.5}$$

$$\delta\widetilde{Q}_{\nu} = g^{\mu\lambda}\delta Q_{\lambda\mu\nu} = \big((D+1)a + 2b\big)\xi_{\nu}. \tag{A.6}$$

At $D = 4$ this is (4.4):

$$\delta Q = (2a + 8b)\xi, \qquad \delta\widetilde{Q} = (5a + 2b)\xi. \tag{A.7}$$

A.3 Linear Ricci coefficients

In a local inertial frame the partial terms of (3.1) with $C^{\lambda}{}_{\mu\nu} = a\delta^{\lambda}{}_{\mu}\xi_{\nu} + b\delta^{\lambda}{}_{\nu}\xi_{\mu}$ give, after Ricci contraction $R_{\alpha\beta} = R^{\lambda}{}_{\alpha\lambda\beta}$,

$$\delta R_{\alpha\beta}\big|_{\partial} = b\,\partial_{\alpha}\xi_{\beta} + (-3a - b)\,\partial_{\beta}\xi_{\alpha}. \tag{A.8}$$

The quadratic remainder $C{\cdot}C$ in the connection-difference identity is proportional to a in this family and vanishes identically at $a = 0$. The target $\delta R_{\mu\nu} = -2\partial_{[\mu}\xi_{\nu]}$ requires $b = -1$ and $-3a - b = 1$. Combined with $a - b = 1$ from $\delta T = +3\xi$ and with $\delta A = +3\xi$ imposed, the unique solution is $a = 0$, $b = -1$.

A.4 Direct curvature variation at $a = 0$, $b = -1$

Set $C^{\lambda}{}_{\mu\nu} = -\delta^{\lambda}{}_{\nu}\xi_{\mu}$. Mixed Γ–C terms cancel pairwise and $C{\cdot}C$ cancels by symmetry of $\xi_{\mu}\xi_{\nu}$. The remainder is

$$\delta R^{\lambda}{}_{\rho\mu\nu} = \partial_{\mu} C^{\lambda}{}_{\nu\rho} - \partial_{\nu} C^{\lambda}{}_{\mu\rho} = -\delta^{\lambda}{}_{\rho}(\partial_{\mu}\xi_{\nu} - \partial_{\nu}\xi_{\mu}), \tag{A.9}$$

hence $\delta R_{\rho\nu} = -2\partial_{[\rho}\xi_{\nu]}$ and $\delta R = 0$, which is (4.10).

A.5 Alternative first-index shift under the present targets

The pair $(a, b) = (1, 0)$ yields $\delta T = +3\xi$ but $\delta(Q, \widetilde{Q}) = (2, 5)\xi$ and $b = 0$ in (4.5), so the Ricci target fails. It is a standard first-kind MAG writing elsewhere. Under the present targets it is not selected by the two-parameter family; it is not declared incorrect as a general convention.

B Trace-distortion inversion

B.1 One-form ansatz

On the trace-only sector, a metric-covariant tensor linear in a one-form v_μ has the form

$$N^\lambda{}_{\mu\nu}[v] = a\,\delta^\lambda{}_\mu v_\nu + b\,\delta^\lambda{}_\nu v_\mu + c\,g_{\mu\nu}v^\lambda. \tag{B.1}$$

An independent triple (a, b, c) is used for each of T, Q, and $\widetilde{Q}$.

B.2 Trace map and inverse

Let (t, q, r) be the reconstructed $(T, Q, \widetilde{Q})$ coefficients produced by one such v. Direct contraction gives

$$t = (D-1)(a-b), \qquad q = 2(a + Db + c), \qquad r = (D+1)(a+c) + 2b. \tag{B.2}$$

With $L = (D-1)(D+2) = D^2 + D - 2$, inversion yields

$$\begin{aligned} a &= \frac{t}{D-1} + \frac{(D+1)q}{2L} - \frac{r}{L}, \\ b &= \frac{(D+1)q}{2L} - \frac{r}{L}, \\ c &= -\frac{t}{D-1} - \frac{(D+3)q}{2L} + \frac{(D+1)r}{L}. \end{aligned} \tag{B.3}$$

B.3 Four-dimensional distortion

At $D = 4$, applying the inverse separately to unit T, Q, and $\widetilde{Q}$ data produces (4.13). No other irrep enters.

B.4 Horizontal change of basis

In the ordered raw basis $(T, Q, \widetilde{Q}, A)$ and horizontal basis (t, q, u, V),

$$\begin{pmatrix} t \\ q \\ u \\ V \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 8/3 \\ 0 & 0 & 1 & 2/3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} T \\ Q \\ \widetilde{Q} \\ A \end{pmatrix}. \tag{B.4}$$

The determinant is 1. The inverse is (4.15). The raw projective vector $(3, -8, -2, 3)$ is sent to $(0, 0, 0, 3)$. Any $G \in \mathrm{GL}(3)$ acting on (q, u, t) multiplies $\det H_\perp$ by $(\det G)^2$ and leaves the regular locus $f_1 f_2 = 0$ unchanged.

C Full curvature expansion

C.1 Connection-difference identity

With $\Gamma = \{\} + N$, the scalar curvature splits into R_{LC}, a divergence built from traces of N, and a quadratic form in N. In components, writing N for N,

$$R - R_{\mathrm{LC}} = \bar{\nabla}_\mu\big(N^\mu{}_\lambda{}^\lambda - N^\lambda{}_\lambda{}^\mu\big) + g^{\rho\nu}\big(N^\lambda{}_{\lambda\kappa}N^\kappa{}_{\nu\rho} - N^\lambda{}_{\nu\kappa}N^\kappa{}_{\lambda\rho}\big), \tag{C.1}$$

up to the standard rearrangement that identifies the displayed divergence with $\bar{\nabla}_\mu W^\mu$ after the trace-only substitution. (The identity is the contraction of (3.1) and does not use the projective law.)

C.2 Substitution of the trace distortion

Insert (4.13) and contract at $D = 4$. The divergence coefficients are

$$N^\mu{}_\lambda{}^\lambda - N^\lambda{}_\lambda{}^\mu = -2T^\mu - Q^\mu + \widetilde{Q}^\mu = W^\mu. \tag{C.2}$$

The quadratic polynomial is bilinear in $(T, Q, \widetilde{Q})$. Collecting T^2, Q^2, $\widetilde{Q}^2$, $T\cdot Q$, $T\cdot\widetilde{Q}$, and $Q\cdot\widetilde{Q}$ produces exactly the six coefficients

$$\big(-\tfrac{2}{3},\ -\tfrac{11}{72},\ +\tfrac{1}{18},\ -\tfrac{2}{3},\ +\tfrac{2}{3},\ +\tfrac{2}{9}\big), \tag{C.3}$$

which is (5.1). There is no seventh trace structure and no remainder in this sector.

C.3 Horizontal independence of V

Substitute (4.15). Then

$$W = -2(t+V) - \big(q - \tfrac{8}{3}V\big) + \big(u - \tfrac{2}{3}V\big) = -2t - q + u, \tag{C.4}$$

$$T^2 = (t+V)^2, \quad Q^2 = \big(q - \tfrac{8}{3}V\big)^2, \quad \widetilde{Q}^2 = \big(u - \tfrac{2}{3}V\big)^2, \tag{C.5}$$

and likewise for the three cross terms. Expanding and using the six coefficients, every monomial containing V cancels. The quadratic form in (t, q, u) is identical to that in $(T, Q, \widetilde{Q})$. This is the second, trace-polynomial, proof of $\delta R = 0$.

C.4 Hessian H_0

Let $\mathsf{Q}(q, u, t)$ denote the quadratic polynomial of (5.1) with $(T, Q, \widetilde{Q}) \mapsto (q, u, t)$. The Hessian matrix with respect to (q, u, t), treating metric contractions as the inner product, is

$$H_0 = \begin{pmatrix} \partial_q\partial_q\mathsf{Q} & \partial_q\partial_u\mathsf{Q} & \partial_q\partial_t\mathsf{Q} \\ \partial_u\partial_q\mathsf{Q} & \partial_u\partial_u\mathsf{Q} & \partial_u\partial_t\mathsf{Q} \\ \partial_t\partial_q\mathsf{Q} & \partial_t\partial_u\mathsf{Q} & \partial_t\partial_t\mathsf{Q} \end{pmatrix} = \begin{pmatrix} -11/36 & 2/9 & -2/3 \\ 2/9 & 1/9 & 2/3 \\ -2/3 & 2/3 & -4/3 \end{pmatrix}, \tag{C.6}$$

because $\partial_q\partial_q(-11/72\,q^2) = -11/36$, $\partial_u\partial_u(1/18\,u^2) = 1/9$, $\partial_t\partial_t(-2/3\,t^2) = -4/3$, $\partial_q\partial_u(2/9\,q\cdot u) = 2/9$, $\partial_q\partial_t(-2/3\,t\cdot q) = -2/3$, and $\partial_u\partial_t(2/3\,t\cdot u) = 2/3$. This is (5.13).

D Palatini block, determinant, and inverse

D.1 Source vector

After (5.5)–(5.6) the linear bulk terms are

$$-(\partial f_1)\cdot W - (\partial f_3)\cdot t = (\partial f_1)\cdot(2t + q - u) - (\partial f_3)\cdot t, \tag{D.1}$$

hence

$$J = \big(\partial f_1,\ -\partial f_1,\ 2\partial f_1 - \partial f_3\big)^{\mathsf{T}}, \tag{D.2}$$

which is (5.14). The $f_2 t^2$ term contributes $\mathrm{diag}(0, 0, 2f_2)$ to the Hessian in the $\frac{1}{2}x^{\mathsf{T}}Hx$ normalization.

D.2 Determinant of P

From (5.15),

$$\det \begin{pmatrix} -11/36 & 2/9 \\ 2/9 & 1/9 \end{pmatrix} = \big(-\tfrac{11}{36}\big)\big(\tfrac{1}{9}\big) - \big(\tfrac{2}{9}\big)^2 = -\tfrac{11}{324} - \tfrac{4}{81} = -\tfrac{11}{324} - \tfrac{16}{324} = -\tfrac{1}{12}. \tag{D.3}$$

Thus $\det P = -f_1^2/12$, which is (5.16).

D.3 Inverse

The adjugate of the numerical matrix is

$$\begin{pmatrix} 1/9 & -2/9 \\ -2/9 & -11/36 \end{pmatrix} \Big/ (-1/12) = \begin{pmatrix} -4/3 & 8/3 \\ 8/3 & 11/3 \end{pmatrix}, \tag{D.4}$$

and scaling by f_1^{-1} yields (5.17). Direct multiplication confirms $PP^{-1} = I_2$ on $f_1 \neq 0$.

D.4 q/u solve at fixed t

Equation (5.18) is $Pq_N = -f_1 c\, t - J_N$ with $c = (-2/3, 2/3)^{\mathsf{T}}$ and $J_N = (\partial f_1, -\partial f_1)^{\mathsf{T}}$. Then

$$-P^{-1}J_N = \big(4r,\ r\big), \tag{D.5}$$

$$-P^{-1}(f_1 c\, t) = \big(-\tfrac{8}{3}t,\ -\tfrac{2}{3}t\big), \tag{D.6}$$

with $r = \partial f_1/f_1$, which is (5.19).

E Schur calculation and complete affine solution

E.1 Schur complement

Partition $H_\perp$ as

$$H_\perp = \begin{pmatrix} P & B \\ B^{\mathsf{T}} & C \end{pmatrix}, \qquad B = f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix}, \qquad C = f_1(-4/3) + 2f_2. \tag{E.1}$$

The Schur complement in the t slot is $S_t = C - B^{\mathsf{T}}P^{-1}B$. Substituting (5.17) gives $B^{\mathsf{T}}P^{-1}B = f_1(-4/3)$, hence $S_t = 2f_2$. The effective t-source after eliminating q_N is

$$J_{t,\mathrm{eff}} = J_t - B^{\mathsf{T}}P^{-1}J_N = (2\partial f_1 - \partial f_3) - 2\partial f_1 = -\partial f_3. \tag{E.2}$$

The t-independent remainder is

$$\mathcal{L}_{\mathrm{const}} = -\tfrac{1}{2}J_N^{\mathsf{T}}P^{-1}J_N = \frac{3(\partial f_1)^2}{2f_1}. \tag{E.3}$$

This is (5.21). Consequently $\Delta_2 = \Delta_3 = 0$ and $\widehat{f_2} = f_2$, $\widehat{f_3} = f_3$.

E.2 Residue solution and reconstruction

$S_t t + J_{t,\mathrm{eff}} = 0$ is $2f_2 t - \partial f_3 = 0$, hence (5.27) on $f_2 \neq 0$. Combined with (5.19) this is (5.29). The raw reconstruction (6.13) follows from (4.15).

E.3 On-shell distortion

Insert $T = s+V$, $Q = 4r - (8/3)(s+V)$, $\widetilde{Q} = r - (2/3)(s+V)$ into (4.13). All terms rearrange into

$$N^{\lambda}{}_{\mu\nu} = \tfrac{1}{2}\big(\delta^{\lambda}{}_{\mu} r_{\nu} + \delta^{\lambda}{}_{\nu} r_{\mu} - g_{\mu\nu} r^{\lambda}\big) - \tfrac{1}{3}\delta^{\lambda}{}_{\nu}(s_{\mu} + V_{\mu}), \tag{E.4}$$

which is (6.14). The V piece is a pure last-index shift: $V \mapsto V + 3\xi$ produces $\delta N^{\lambda}{}_{\mu\nu} = -\delta^{\lambda}{}_{\nu}\xi_{\mu}$.

F Full Hessian kernel

F.1 Raw bulk density

After the admitted integrations by parts, the affine bulk density in raw traces is

$$\mathcal{L} = f_1\, \mathsf{Q}(T, Q, \widetilde{Q}) + (\partial f_1)\cdot(2T + Q - \widetilde{Q}) + f_2(T-A)^2 - (\partial f_3)\cdot(T - A), \tag{F.1}$$

with Q the quadratic polynomial of (5.1). Let $X = (T, Q, \widetilde{Q}, A)$ and let H_{raw} be the Hessian of (F.1) with respect to X.

F.2 Gauge inclusion

Differentiating (F.1) along $(3, -8, -2, 3)$ yields zero identically, because that vector is the projective orbit of Lemma 4.2 and Q together with $W = -2T - Q + \widetilde{Q}$ are invariant, while $T - A$ is invariant. Thus

$$\mathrm{span}\{(3,-8,-2,3)\} \subseteq \ker H_{\mathrm{raw}}. \tag{F.2}$$

F.3 Converse on the regular locus

In the invariant coordinates, $\mathcal{L}$ is independent of V and equals (5.8). Therefore the species matrix is $H_{\mathrm{full}} = \mathrm{diag}(H_{\perp}, 0)$ and $\det H_{\perp} = -f_1^2 f_2/6$. On U_{reg}, $H_{\perp}$ is invertible, so $\ker H_{\mathrm{full}} = \mathrm{span}\{e_V\}$ and $\ker \mathcal{H}_{\mathrm{full}} = \mathrm{im}\, G_x$, which pulls back to the raw fibre kernel (6.11). This is the converse: no additional affine kernel exists in the present trace sector.

F.4 Gauge identity

The Noether identity associated with (4.1) is the vanishing of (6.12). It holds off-shell as an identity among the raw Euler–Lagrange expressions of (F.1), and is equivalent to V-independence of $\mathcal{L}$.

F.5 Singular enlargement

On S2–S4, $\mathrm{rank}_{\mathrm{sp}}(H_{\mathrm{full}})$ drops as in Table 2, so the kernel properly contains the projective tangent. Those extra null directions are not gauge: they are gauge-invariant degeneracies of the quadratic form at vanishing f_1 or f_2, matching the invariant moduli of Appendix I. Selecting a representative among them is a non-gauge restriction. They are excluded from Theorem 7.2 rather than quotiented.

G Explicit QE and EQ substitutions

G.1 Quotient first

On U_{reg}, $H_\perp x + J = 0$ has the unique solution (5.29). Then

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star. \tag{G.1}$$

Using the Schur factorization,

$$\tfrac{1}{2} x_\star^{\mathsf{T}} H_\perp x_\star + J^{\mathsf{T}} x_\star = \mathcal{L}_{\mathrm{const}} + \tfrac{1}{2} S_t t_\star^2 + J_{t,\mathrm{eff}} \cdot t_\star = \frac{3(\partial f_1)^2}{2f_1} + f_2 s^2 - (\partial f_3)\cdot s, \tag{G.2}$$

with $s = \partial f_3/(2f_2)$. The last two terms are

$$f_2\Big(\frac{\partial f_3}{2f_2}\Big)^2 - (\partial f_3)\cdot\frac{\partial f_3}{2f_2} = \frac{(\partial f_3)^2}{4f_2} - \frac{(\partial f_3)^2}{2f_2} = -\frac{(\partial f_3)^2}{4f_2}. \tag{G.3}$$

Hence

$$\mathcal{L}^{\mathrm{QE}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}. \tag{G.4}$$

G.2 Eliminate first

Substitute (6.13) into (F.1). Write $T = s + V$, $Q = 4r - (8/3)(s+V)$, $\widetilde{Q} = r - (2/3)(s+V)$, $A = V$. Every occurrence of V in $\mathsf{Q}(T, Q, \widetilde{Q})$ and in $2T + Q - \widetilde{Q}$ cancels by Appendix C, and $T - A = s$ is independent of V. The remaining expression is a function of (r, s) only. Expanding,

$$\begin{aligned} f_1\,\mathsf{Q}\big|_{V\text{-free}} + (\partial f_1)\cdot(2s + 4r - r) + f_2 s^2 - (\partial f_3)\cdot s \\ = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}, \end{aligned} \tag{G.5}$$

after using $r = \partial f_1/f_1$, $s = \partial f_3/(2f_2)$, and the Palatini evaluation of Q on $(q, u) = (4r - 8s/3,\, r - 2s/3)$. Thus

$$\mathcal{L}^{\mathrm{EQ}}_{\mathrm{aff,on\text{-}shell}} = \frac{3(\partial f_1)^2}{2f_1} - \frac{(\partial f_3)^2}{4f_2}, \tag{G.6}$$

independent of V. The representatives $V = 0$, $V = \mathrm{d}\chi$, and V arbitrary therefore yield the same bulk density.

G.3 Admitted remainder

The two substitutions are performed on the bulk densities after (5.5)–(5.6). They therefore agree modulo

$$\partial_\mu\big(\sqrt{-g}\,(c_1 J_1^\mu + c_3 J_3^\mu)\big). \tag{G.7}$$

Whether this remainder is retained, canceled, or discarded is a variational-boundary convention. Off shell, $\delta J_1 = \delta J_3 = 0$ for projectively inert spectators, so the remainder does not classify proper versus improper transformations (Appendix H).

H Boundary currents and IBP identification

H.1 Currents actually used

The integrations by parts in the proof are only (5.5) and (5.6). They generate

$$J_1^\mu = f_1 W^\mu, \qquad J_3^\mu = f_3 \bar{T}^\mu, \tag{H.1}$$

with $W^\mu = -2T^\mu - Q^\mu + \widetilde{Q}^\mu$ and $\bar{T}^\mu = t^\mu = T^\mu - A^\mu$. No other surface current is invoked.

H.2 Admitted identification

Two bulk functionals are identified when their difference is a total divergence of the linear span of J_1 and J_3. This is a definition of the admitted variational-boundary class, not a claim that all metric-affine boundary terms lie in that span, and not a classification of gauge charges.

H.3 Off-shell projective invariance of the currents

Under the adopted law (4.1),

$$\delta(T, Q, \widetilde{Q}, A) = (3, -8, -2, 3)\xi \tag{H.2}$$

holds off shell. Therefore

$$\delta t = \delta T - \delta A = 0, \tag{H.3}$$

$$\delta W = -2\delta T - \delta Q + \delta\widetilde{Q} = (-6 + 8 - 2)\xi = 0. \tag{H.4}$$

Equivalently, the inverse chart (4.15) gives $W = -2t - q + u$ identically, with no V remainder. If the spectator coefficients f_1 and f_3 are projectively inert,

$$\delta J_1^\mu = 0, \qquad \delta J_3^\mu = 0 \tag{H.5}$$

off shell. Consequently

$$n_\mu \delta(J_1^\mu + J_3^\mu) = 0 \tag{H.6}$$

holds for every projective transformation of this class. That identity cannot distinguish proper from improper projective transformations.

Two distinct issues must therefore be kept separate.

A. IBP / variational boundary treatment. Whether the displayed total divergence of J_1 and J_3 is retained, canceled, or discarded is a choice of boundary data for the variational problem. Theorem 6.3 identifies QE and EQ as spacetime-local bulk functionals modulo that divergence.

B. Gauge charges / improper transformations. A Hamiltonian or covariant-phase-space analysis of charges carried by ξ is a separate question. It is not established here, and is not attempted.

H.4 Tests

Test	Content	Inside Theorem 7.2
A	bulk QE = EQ modulo J_1, J_3	yes
B	retain, cancel, or discard the displayed IBP divergence	boundary data, not a hypothesis
C	Hamiltonian or covariant-phase-space charge of ξ	no

No global boundary classification, charge algebra, or topological sector is claimed. The reduced functional (7.1) is not asserted to be a complete initial-boundary-value action without further boundary data.

H.5 Local versus global vocabulary

Claim	Scope
Palatini inverse	field-space local, on U_{reg}
fibre kernel equals $\operatorname{im} G_x$	field-space local, on U_{reg}
connection solve	pointwise algebraic
orbit uniqueness	field-space local and spacetime local, modulo (4.1)
QE = EQ	spacetime-local bulk functionals modulo J_1, J_3
reduced functional	spacetime-local bulk plus admitted IBP remainder
global gauge slice	not claimed
projective charges of ξ	not claimed
all gauges / all connections	not claimed

I Singular strata and counterexamples

The following examples show that the hypotheses of Theorem 7.2 are necessary. They are not solutions of the singular dynamics. All statements start from $H_\perp x + J = 0$ with $x = (q, u, t)$ and $J = (\partial f_1, -\partial f_1, 2\partial f_1 - \partial f_3)^{\mathsf{T}}$, without dividing by f_1 or f_2.

S1, $f_1 \neq 0$, $f_2 \neq 0$. $\operatorname{rank}_{\mathrm{sp}}(P) = 2$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 3$, $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$. At $D = 4$ the fibre values are $\operatorname{rank} \mathcal{H}_{\mathrm{full}} = 12$ and $\dim \ker \mathcal{H}_{\mathrm{full}} = 4$. The source J always lies in the image, and the unique horizontal solution is (5.29). The regular-locus theorem is unchanged.

S2, $f_1 = 0$, $f_2 \neq 0$. $P = 0$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 1$, $\operatorname{rank}_{\mathrm{sp}}(H_{\mathrm{full}}) = 1$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 3$. At $D = 4$ the fibre values are rank 4 and nullity 12. The system collapses to $\partial f_1 = 0$ together with $2 f_2\, t = \partial f_3$. Compatibility therefore requires $\partial f_1 = 0$. If $\partial f_1 \neq 0$, there is no stationary horizontal solution. If $\partial f_1 = 0$, then $t = \partial f_3/(2 f_2)$ while q and u remain invariant algebraic moduli. The extra kernel directions are these horizontal moduli together with the vertical gauge V; they are not additional projective gauge.

S3, $f_1 \neq 0$, $f_2 = 0$. $\operatorname{rank}_{\mathrm{sp}}(P) = 2$, $\operatorname{rank}_{\mathrm{sp}}(H_\perp) = 2$, $\operatorname{null}_{\mathrm{sp}}(H_{\mathrm{full}}) = 2$. At $D = 4$ the fibre values are rank 8 and nullity 8. The Palatini block still determines (q, u) as functions of t. Substituting into the residue row yields the identity $-\partial f_3 = 0$ independently of t. Compatibility therefore requires $\partial f_3 = 0$. If $\partial f_3 \neq 0$, there is no stationary solution. If $\partial f_3 = 0$, then t is a gauge-invariant modulus and (q, u) are determined by t. The extra kernel is this invariant t-direction, not additional gauge.

S4, $f_1 = f_2 = 0$. $H_\perp = 0$, species kernel dimension four, fibre kernel dimension 16 at $D = 4$. Compatibility is $J = 0$, i.e. $\partial f_1 = 0$ and $\partial f_3 = 0$. If either fails, there is no stationary solution. If both hold, (q, u, t) are horizontal moduli and V remains vertical gauge.

Alternative first-index shift under the present targets. The pair $(a, b) = (1, 0)$ induces the trace map $(3, 2, 5)\xi$. It fails the Ricci target (4.6) and is not selected by Proposition 4.1.

IBP remainder versus charges. The bulk density remains projectively invariant. Off shell, $\delta J_1 = \delta J_3 = 0$ for inert spectators. The condition $n_\mu \delta(J_1^\mu + J_3^\mu) = 0$ therefore cannot classify proper versus improper transformations. Test C of Appendix H lies outside the theorem.

Deleted irreps as unknowns. Axial torsion, tensor torsion, and traceless nonmetricity are not coordinates of H_{full}. Treating them as solved variables changes the domain rather than enlarging the kernel computation.

No additional algebraic factor of $\det H_\perp$ appears. The regular proof is not extended through S2–S4.

J Minimal deformation control of the residue fingerprint

The following deformations test the residue fingerprint; the horizontal system defined by (3.7) is left unchanged. A local deformation is called *symmetry-compatible* when it is a diffeomorphism scalar constructed from the projectively invariant horizontal one-forms, at the same algebraic and derivative order as the undeformed quadratic density, and independent of the vertical coordinate V.

Write $x = (q, u, t)$ and isolate the Palatini block P, mixing B, and residue coefficient C in the $\frac{1}{2}x^{\mathsf{T}}Hx$ normalization,

$$B = f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix}, \qquad C = f_1\big(-\tfrac{4}{3}\big) + 2f_2, \tag{J.1}$$

so that the t-Schur coefficient is $S_t = C - B^T P^{-1} B$. The undeformed value is $S_t(0,0) = 2f_2$, equivalently $\Delta_2 = 0$.

Consider the nearby symmetry-compatible quadratic deformation

$$\delta L_{\text{quad}} = \alpha\, q \cdot t + \beta\, u \cdot t, \tag{J.2}$$

with independent symbolic coefficients α, β. This shifts $B \mapsto B + \delta B$ with $\delta B = (\alpha, \beta)^{\mathsf{T}}$. Then

$$S_t(\alpha, \beta) = C - (B + \delta B)^T P^{-1} (B + \delta B). \tag{J.3}$$

Using (5.17) one finds $B^T P^{-1} \delta B = \frac{8}{3}\alpha + \frac{2}{3}\beta$ and

$$\delta B^T P^{-1} \delta B = f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.4}$$

Hence

$$\delta S_t := S_t(\alpha, \beta) - S_t(0,0) = -\tfrac{16}{3}\alpha - \tfrac{4}{3}\beta - f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.5}$$

For each $f_1 \neq 0$, the zero locus of (J.5) is a plane conic of measure zero in the (α, β) plane. Off that conic, $\delta S_t \neq 0$, so Δ_2 is corrected. In particular, the line $\beta = -4\alpha$ kills the linear part and yields $\delta S_t = -36\alpha^2/f_1$, which vanishes only at $\alpha = 0$.

The undeformed t-source Schur correction is $J_{t,\text{eff}} = -\partial f_3$, equivalently $\Delta_3 = 0$. With (J.2) one obtains the extra term

$$J_{t,\text{eff}}(\alpha, \beta) = -\partial f_3 + \frac{\partial f_1}{f_1}(4\alpha + \beta). \tag{J.6}$$

Thus Δ_3 remains zero if and only if $\beta = -4\alpha$. Combined with (J.5), both $\Delta_2 = \Delta_3 = 0$ hold simultaneously only at $(\alpha, \beta) = (0, 0)$.

A minimal symmetry-compatible linear-source deformation $J_N \mapsto J_N + (\varepsilon_q, \varepsilon_u)$ produces, at $\delta B = 0$,

$$\delta J_{t,\text{eff}} = -\big(\tfrac{8}{3}\varepsilon_q + \tfrac{2}{3}\varepsilon_u\big). \tag{J.7}$$

This vanishes if and only if $\varepsilon_u = -4\varepsilon_q$. A generic source deformation therefore yields a nonzero t-source Schur correction.

The fingerprint $\Delta_2 = \Delta_3 = 0$ is explicitly non-generic within the stated deformation family. These diagnostic deformations do not modify the defining equations of the model.

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